

An infinite finitely presented residually finite 2-group

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Abstract

We prove that the infinite, ordinarily finitely presented periodic Steinberg group $\Gamma = \text{St}_{12}(R)$ of a companion paper is residually finite. Its finite-index subgroup $G = \ker(\Gamma \rightarrow \text{St}_{12}(\mathbb{F}_2))$ is infinite, ordinarily finitely presented, and residually finite, and every element of G has finite 2-power order. The orders of its elements are unbounded. Phases of long words in the companion’s graded algebra allow finite degree truncations to detect all elements of Γ , including the central kernel.

1 Introduction

A group is *periodic* if each of its elements has finite order. It is *residually finite* if, for every nonidentity element g , there is a homomorphism to a finite group whose value at g is nonidentity. We use *2-group* to mean a group in which every element has finite 2-power order; the group itself need not be finite. An *ordinary finite presentation* is a quotient of a free group on finitely many generators by the normal closure of finitely many words. We prove the following.

Theorem 1.1. *There exists an infinite residually finite group with an ordinary finite presentation in which every element has finite 2-power order.*

The group is an abstract discrete group. Finite presentation is meant in that category, rather than in a variety of groups or in the category of pro-2 groups. The orders of its elements are necessarily unbounded.

History and relation to earlier constructions

Burnside’s finiteness questions [2] ask how finite-order conditions constrain a group with finitely many generators. Golod’s construction [3, Example 2], based on the Golod–Shafarevich dimension argument [4], already produces infinite finitely generated residually finite p -groups. Its algebraic mechanism is particularly relevant here: nilpotence gives finite p -power orders, while a grading separates elements in finite quotients. Grigorchuk later gave a concrete infinite finitely generated residually finite 2-group [5]; his original theorem also establishes that this group is not finitely presented. Thus residual finiteness and fixed-prime torsion can coexist with finite generation. The additional requirement in Theorem 1.1 is an ordinary finite presentation.

The question whether an infinite periodic group can have an ordinary finite presentation is recorded in [12, p. 45] and again in [6, Section 2.2(IX)]. The torsion-by-cyclic groups of [12, Theorem 1.1] have an infinite cyclic quotient. They establish a different finite-presentation result and are not themselves periodic. The companion paper [13] constructs an infinite finitely presented periodic group using a graded algebra with a matrix-nil positive-degree ideal. The present paper

strengthens that construction by proving residual finiteness and passing to a finite-index subgroup with 2-power torsion. The companion's algebra and finite local-rule hierarchy are essential inputs; we state their precise properties in Section 2.

A bound on all element orders would change the problem. Novikov and Adian constructed infinite finitely generated groups of bounded exponent [11]. In the residually finite setting, however, Zel'manov's solution of the restricted Burnside problem for 2-groups [20, Theorem 2] implies that a finitely generated residually finite group of exponent dividing 2^a is finite. Consequently the group in Theorem 1.1 has unbounded exponent. Its finite quotients are nevertheless all 2-groups: a finite image has only elements of 2-power order, and Cauchy's theorem excludes every odd prime divisor of its order.

The proof and its main ingredients

The companion supplies an infinite finitely presented graded algebra

$$R = \mathbb{F}_2 \oplus I, \quad I = \bigoplus_{d \geq 1} R_d,$$

such that every matrix in $M_n(I)$ is nilpotent, for every finite n . Each homogeneous component R_d is finite-dimensional. The elementary matrix group $E_n(R)$ is generated by the matrices $e_{ij}(a)$ obtained by adding $a \in R$ to an off-diagonal position of the identity. The Steinberg group $St_n(R)$ is generated by symbols $x_{ij}(a)$ subject to the elementary addition and nonopposite-root commutator relations; these are written explicitly in Section 3. There is a natural map

$$St_n(R) \longrightarrow E_n(R), \quad x_{ij}(a) \longmapsto e_{ij}(a).$$

Finite presentability of Steinberg groups in rank at least four over finitely presented unital rings is due to Krstić and McCool [8, Theorem 3]; an adapted proof is also given in [13, Theorem 4.1]. We take $n = 12$ and eventually use the kernel of reduction from $St_{12}(R)$ to $St_{12}(\mathbb{F}_2)$.

The degree truncations $R/R_{\geq d}$, where $R_{\geq d} = \bigoplus_{b \geq d} R_b$, are finite rings. They immediately detect nonidentity matrices over R . The main difficulty is to detect elements of the Steinberg kernel as well. We first work in the stable group $St(R) = \varinjlim_n St_n(R)$, put $E(R) = \varinjlim_n E_n(R)$, and write

$$J(R) = \ker(St(R) \longrightarrow E(R)).$$

This kernel is central. We first prove detection in the stable group, where a finite calculation may always use additional coordinates. We return to fixed rank afterward.

Two calculations of one cyclic substitution provide the detection argument. If $z \in J(R)$, the grading replaces each homogeneous root entry $a \in R_d$ by at^d . Substituting the cyclic shift matrix C_L for t and expanding into ordinary roots gives an element $W_z(C_L) \in J(R)$. This is a matrix realization of a graded transfer. Weibel's work on graded rings [18] develops the corresponding transfer operations and their eventual vanishing; it also implies p -primary torsion in relative graded K -groups in characteristic p . Following the elementary Steinberg treatment in [13], Section 4 proves directly that

$$W_z(C_L) = 1 \quad \text{for all sufficiently large } L$$

whenever z reduces to 1 over $\mathbb{F}_2$. Taking L to be a power of two also shows that every such z has 2-power order.

The second calculation uses additional structure of this particular ring. Its finite local rules give a basis of nonzero word classes. For arbitrarily large odd integers L , each sufficiently long

nonzero word has a start phase $\sigma(w) \in \mathbb{Z}/L$ and an end phase $\tau(w) = \sigma(w) + |w|$. These phases depend only on the word class, and a nonzero product must match end phase to start phase. Section 2 derives the phases from the hierarchy's bounded token decompositions.

If z is missed by all degree truncations, it can be represented using roots whose entries are arbitrarily long words. The phases allow us to move the endpoints of each such root to coordinates labeled by its start and end phases, without changing z . Establishing this identity requires control of opposite-root commutators in the Steinberg group, as well as a subgroup on which the matrix map is injective. Sections 5 and 6 prove these facts and obtain, for the long-entry representative,

$$W_z(C_L) = z^L.$$

We first choose an odd L for which the first calculation vanishes, and then find a representative long enough for the second calculation. Thus $z^L = 1$. Since z also has 2-power order, it is the identity. This proves separation by finite degree truncations, the additional step beyond torsion supplied by graded transfers alone.

To pass back to twelve coordinates, we need the injectivity of $\text{St}_{12}(R) \rightarrow \text{St}(R)$. Stability over the local ring R is covered by the classical theorem of Suslin and Tulenbaev [16]. Section 7 gives a direct integral homology proof using ordered frames and universal central extensions of their stabilizers. This supplies the bridge from stable detection to the finitely presented group.

All arguments establishing these phase, stabilization, and detection claims are included below. We quote from the companion the algebra construction and the finite-rank elementary facts needed to apply them, with their hypotheses stated at the points of use. The final passage to finite groups and to the subgroup of 2-power torsion is carried out in Section 8.

2 The algebra and phases of long words

The ring used below has two relevant features. Its positive-degree matrices are nilpotent, which will give torsion, and its long nonzero words carry compatible positions modulo arbitrarily large odd integers, which will give detection by finite quotients. The first feature is an input from the companion paper. We derive the second from the word hierarchy in that paper.

An ideal I of a unital associative algebra is *matrix-nil* if every square matrix of every finite size with entries in I is nilpotent. All algebras in this paper are over $\mathbb{F}_2$.

Proposition 2.1 (The companion algebra). *There is an infinite finitely presented unital associative algebra*

$$R = \bigoplus_{d \geq 0} R_d = \mathbb{F}_2 \oplus I, \quad R_0 = \mathbb{F}_2, \quad I = \bigoplus_{d \geq 1} R_d,$$

such that every R_d is finite-dimensional and I is matrix-nil. This algebra has the word presentation and hierarchy specified in Proposition 2.2.

This is the algebra constructed in [13, Theorem 3.7]. We now give the precise part of its structure that we use. In a presentation by words, a *replacement* is a relation $v = v'$ between nonempty words of equal length, usable in either direction inside any word. A *zero declaration* is a relation $v = 0$ for a nonempty word v . The length of a replacement is the common length of its two sides. For a word w , write $|w|$ for its length and $[w]$ for its image in the relevant algebra.

Proposition 2.2 (The companion hierarchy). *The algebra R in Proposition 2.1 is the first algebra A_0 of a sequence $(A_j)_{j \geq 0}$ with the following properties. There are positive integers c, k, h_0, D , with $D > 10(c+1)(k+1)$, and*

$$h_j = h_0 D^j, \quad L_j = 2 \cdot 2^{h_j} + 1.$$

For each $j \geq 0$:

- (i) A_j is presented by a finite alphabet Σ_j , finitely many replacements, and finitely many zero declarations, all of length at most k . It is graded by word length.
- (ii) There is a substitution E_j sending each letter of Σ_{j+1} to a word of length L_j over Σ_j . Extended by concatenation, it induces a unital algebra homomorphism $A_{j+1} \rightarrow A_j$.
- (iii) A specified subset of Σ_j gives cuts immediately before its letters. There are integers $B_j, T_j \geq 0$ and a finite set of words $\mathcal{T}_j$ such that every nonzero word w of length at least T_j has the unique decomposition

$$w = p_w t_1 \cdots t_r s_w, \quad |p_w|, |s_w| \leq B_j, \quad t_i \in \mathcal{T}_j. \quad (2.1)$$

Here p_w precedes the first cut, s_w begins at the last cut, and the t_i are exactly the intervals between consecutive cuts. A cut is counted only if its following letter occurs in the word. For each $t \in \mathcal{T}_j$ there is a nonempty word u_t over Σ_{j+1} such that

$$1 \leq |u_t| \leq k, \quad |t| = L_j |u_t|, \quad [t] = [E_j(u_t)] \quad \text{in } A_j. \quad (2.2)$$

These are the applicable conclusions of [13, Proposition 2.2 and the choice of parameters in Section 3]. In particular, the cuts in (iii) are recognized from individual letters; restricting to a subword cannot create or move a cut. We will not need arbitrary concatenations of tokens to be nonzero.

We recall the elementary word-basis argument [13, Lemma 2.1], since it will also identify exactly which equalities of long words must preserve their phases. Form a graph whose vertices are all words, including the empty word, and whose edges are replacements in arbitrary contexts. Quotienting the free algebra by the replacement relations leaves one basis vector for each connected component: the defining ideal is precisely the span of differences of words in a common component. Imposing the zero declarations then kills exactly the components containing a word with a zero-declared subword. Thus, in every A_j , the nonzero replacement classes form a basis, and

$$[w] = 0 \iff w \text{ can be replaced by a word containing a zero-declared subword.} \quad (2.3)$$

In particular, a subword of a nonzero word is nonzero. The product of two basis classes is either zero or the class of their concatenation. We will often use the same symbol for a nonzero word and its basis class when it occurs as an algebra entry.

The finite alphabet also shows directly that each homogeneous part R_d is finite-dimensional. Write

$$R_{\geq d} = \bigoplus_{e \geq d} R_e.$$

For $d \geq 1$ this is a two-sided ideal, and $R/R_{\geq d}$ is a finite ring. The remaining task in this Section is to extract compatible modular positions from the hierarchy.

Lemma 2.3 (Phases of long words). *For every integer $d \geq 0$, put*

$$L = L_0 L_1 \cdots L_d.$$

There is an integer $N \geq 1$ such that every nonzero word w in R of length at least N has a start phase $\sigma(w) \in \mathbb{Z}/L\mathbb{Z}$ and an end phase

$$\tau(w) = \sigma(w) + |w| \in \mathbb{Z}/L\mathbb{Z}$$

with the following properties:

- (i) Both phases depend only on the basis class of w .
- (ii) If v is a contiguous subword of w , has length at least N , and starts at offset b from the beginning of w , then

$$\sigma(v) = \sigma(w) + b. \quad (2.4)$$

The integers L so obtained are odd and arbitrarily large.

Proof. Fix d . First we construct phases for literal nonzero words, without claiming invariance under replacements. We work backward through the levels $j = d, d-1, \dots, 0$. At level $d+1$, use the trivial phase modulo the empty product 1, with threshold 1. Suppose that phases with the subword property have been constructed at level $j+1$ modulo

$$L' = L_{j+1} \cdots L_d,$$

above a threshold $n_{j+1} \geq 1$.

For every token $t \in \mathcal{T}_j$, choose one word u_t satisfying (2.2), and use this same choice at every occurrence of t . For the decomposition (2.1), set

$$U(w) = u_{t_1} \cdots u_{t_r}.$$

Choose

$$n_j \geq \max\{1, T_j, 2B_j + L_j n_{j+1}\}. \quad (2.5)$$

Then every nonzero word w of length at least n_j has such a decomposition and

$$|U(w)| = \frac{|w| - |p_w| - |s_w|}{L_j} \geq n_{j+1}.$$

Moreover, $U(w)$ is nonzero in A_{j+1} . Indeed, the substitution homomorphism and the token equalities give

$$[w] = [p_w] [E_j(U(w))] [s_w],$$

so $[U(w)] = 0$ would imply $[w] = 0$. Define the literal start phase by

$$\sigma_j(w) = L_j \sigma_{j+1}(U(w)) - |p_w| \pmod{L_j L'}. \quad (2.6)$$

Multiplication by L_j makes this independent of the representative chosen for the residue $\sigma_{j+1}(U(w))$ modulo L' .

We verify the subword property at level j . Let v be a subword of w of length at least n_j , starting at offset b . Its cuts are exactly the cuts of w whose following letters lie in v . Consequently, its complete tokens are a consecutive subsequence of the complete tokens of w . Our fixed token choices make $U(v)$ a literal subword of $U(w)$. If a is its upper-level offset, the lower-level distance to its first complete token can be computed in two ways:

$$b + |p_v| = |p_w| + L_j a. \quad (2.7)$$

Both upper words meet the upper-level threshold. Applying the inductive subword property and then (2.7) yields

$$\begin{aligned} \sigma_j(v) &= L_j(\sigma_{j+1}(U(w)) + a) - |p_v| \\ &= L_j \sigma_{j+1}(U(w)) - |p_w| + b = \sigma_j(w) + b \pmod{L_j L'}. \end{aligned}$$

This completes the backward induction.

At level zero we now have literal phases modulo L , with the subword property for lengths at least n_0 . Only invariance under replacements remains. Choose

$$N = 2n_0 + k.$$

Consider one replacement inside a nonzero word of length at least N . It affects at most k letters, so an unchanged prefix or an unchanged suffix has length at least n_0 . That common subword has the same offset before and after the replacement, because the replacement preserves length. Applying the literal subword property to both words shows that their phases agree. Every intermediate word in a nonzero replacement class is nonzero. Chaining these equalities proves that the start phase is constant on each class. Word length is constant on each class as well, so the end phase is also invariant. The subword property already proved continues to hold after increasing the threshold from n_0 to N . Finally, every L_j is an odd integer greater than one, so the products $L_0 \cdots L_d$ are odd and unbounded. $\square$

Corollary 2.4. *Fix L, N, σ, τ from Lemma 2.3. If u, v are nonzero basis words of lengths at least N , then*

$$\tau(u) \neq \sigma(v) \implies uv = 0.$$

If $uv \neq 0$, its phases satisfy $\sigma(uv) = \sigma(u)$ and $\tau(uv) = \tau(v)$.

Proof. If the concatenation is nonzero, apply (2.4) to its prefix u and its suffix v . This gives $\sigma(uv) = \sigma(u)$ and $\sigma(v) = \sigma(uv) + |u| = \tau(u)$; adding $|v|$ gives the end-phase identity. $\square$

3 Steinberg groups and faithful strips

We next introduce the group that records elementary matrix operations before imposing any relations beyond the Steinberg relations. Its kernel over matrices is the part that finite degree truncations must detect. All algebras in this section are unital associative $\mathbb{F}_2$ -algebras.

For such an algebra B and $n \geq 3$, the group $\text{St}_n(B)$ has generators $x_{ij}(a)$, where $a \in B$ and $1 \leq i, j \leq n$ with $i \neq j$, and relations

$$x_{ij}(a)x_{ij}(b) = x_{ij}(a + b), \tag{3.1}$$

$$[x_{ij}(a), x_{lt}(b)] = 1 \quad (j \neq l, i \neq t), \tag{3.2}$$

$$[x_{ij}(a), x_{jt}(b)] = x_{it}(ab) \quad (i, j, t \text{ distinct}). \tag{3.3}$$

Our convention is $[X, Y] = XYX^{-1}Y^{-1}$. In particular, $x_{ij}(0) = 1$ and each root element has square 1. The elementary matrix $e_{ij}(a)$ is the identity with a added in position (i, j) . These matrices satisfy (3.1)–(3.3), giving an epimorphism

$$\pi_n : \text{St}_n(B) \longrightarrow \text{E}_n(B) := \langle e_{ij}(a) : a \in B, i \neq j \rangle.$$

Write $J_n(B) = \ker \pi_n$. We use the same presentation with a countable index set to define $\text{St}(B)$, and put

$$\text{E}(B) = \varinjlim_n \text{E}_n(B), \quad J(B) = \ker(\text{St}(B) \longrightarrow \text{E}(B)).$$

Thus $\text{St}(B) = \varinjlim_n \text{St}_n(B)$: each word and each finite derivation uses only finitely many indices. This is a direct limit of discrete groups. No completion is involved. The fixed-rank injectivity needed for R will be proved separately in Section 7.

The presentation and its relation to algebraic K -theory are classical; see [10, 19]. We will use the following elementary consequences repeatedly. They also explain when an identity of matrices does imply the corresponding identity in the Steinberg group.

Lemma 3.1 (Faithful strips). *Let S, T be disjoint finite sets of coordinates. The subgroup generated by $x_{ij}(a)$ with $i \in S, j \in T$ maps isomorphically onto the additive group of matrices supported in $S \times T$ under $u \mapsto \pi(u) - I$. A word supported on S normalizes this subgroup and acts on its entries by left multiplication by its matrix on S . On the reverse strip $T \times S$, its action is right multiplication by the inverse matrix on S .*

Proof. All roots in $S \times T$ commute by (3.2). Root addition collects a word into one factor per coordinate pair. The corresponding entry matrices have pairwise zero products, so the matrix image reads each collected coefficient directly. This proves injectivity and identifies the group law with addition.

For one root supported on S , conjugation on an outgoing strip follows from (3.3): it performs the corresponding elementary row addition. The same relations on the reverse strip perform the inverse elementary column operation. Iterating over the factors of a word gives the stated left and right actions, with their prescribed order over the possibly noncommutative algebra B . $\square$

Lemma 3.2 (Stable centrality). *For every B , the group $J(B)$ is central in $\text{St}(B)$.*

Proof. Let $z \in J(B)$ and choose a finite set S containing the support of a representative of z . To show that z commutes with a specified root, enlarge S to contain both indices of that root. Choose $h \notin S$. By Lemma 3.1, z acts trivially on the strips $S \times \{h\}$ and $\{h\} \times S$, since its matrix image is the identity. For $i, j \in S$ with $i \neq j$, the identity

$$x_{ij}(a) = [x_{ih}(a), x_{hj}(1)]$$

therefore implies that z commutes with $x_{ij}(a)$. These roots generate the group. $\square$

Lemma 3.3 (Coordinate permutations). *In characteristic two, every permutation of a finite set of coordinates is induced on roots by conjugation in the Steinberg group after adjoining an outside coordinate if necessary. In particular, relabeling the finitely many coordinates in a representative of $z \in J(B)$ leaves the element z unchanged in $\text{St}(B)$.*

Proof. For a transposition of coordinates a, b , use

$$p = x_{ab}(1)x_{ba}(1)x_{ab}(1).$$

On a column strip from $\{a, b\}$ to an outside coordinate, its three elementary row additions exchange the two entries over $\mathbb{F}_2$. Lemma 3.1 gives this as an exact group action. The reverse strip is exchanged in the same way. Roots with both indices outside $\{a, b\}$ commute with p . The remaining two root positions can be written, with $c \notin \{a, b\}$, as

$$x_{ab}(u) = [x_{ac}(u), x_{cb}(1)], \quad x_{ba}(u) = [x_{bc}(u), x_{ca}(1)].$$

The strip actions exchange these as well. Products of transpositions give arbitrary finite permutations. Any injective relabeling of a finite set extends to a permutation of a larger finite set, so the last assertion follows from Lemma 3.2. $\square$

This proof of Lemma 3.3 is the argument of [13, Lemma 6.3]; its exact action on roots, rather than only on their matrix images, will matter below.

Lemma 3.4 (A commutator identity). *In any group, let $A = [X, Y]$ and $B' = [Y, Z]$. If*

$$[X, Z] = [A, X] = [B', Z] = [A, B'] = 1,$$

then $[A, Z] = [X, B']$. This common commutator commutes with A, Z, X, B' .

Proof. The subgroups $H = \langle A, Z \rangle$ and $K = \langle X, B' \rangle$ commute elementwise by the four hypotheses. Conjugating $[X, Z] = 1$ by Y gives

$$[A^{-1}X, B'Z] = 1.$$

Separating the commuting H and K factors yields $[A^{-1}, Z][X, B'] = 1$. Thus $[X, B'] = [A^{-1}, Z]^{-1}$ lies in $H \cap K$, which is central in both H and K . Centrality in H then gives $[A, Z] = [A^{-1}, Z]^{-1} = [X, B']$. $\square$

Finally, $\text{St}_n(B)$ and $E_n(B)$ are perfect for $n \geq 3$: each generating root is the commutator in (3.3) with one coefficient equal to 1. If B is infinite, both groups are infinite, since distinct coefficients at a fixed root position give distinct elementary matrices.

4 Cyclic shifts and the relative central kernel

We now study elements of $J(R)$ whose reduction to $\text{St}(\mathbb{F}_2)$ is trivial. The grading lets us evaluate such an element at a finite numerical matrix. We shall prove that evaluation at a sufficiently long cyclic shift is trivial, and that evaluation at a cyclic shift of 2-power size is a power of the original element. These two identities imply 2-primary torsion in the relative central kernel. Later we will compare the first identity with an odd power.

The matrix substitutions are elementary versions of the graded transfer operations of Weibel [18, Section 1]. At the level of stable algebraic K -theory, the relative torsion conclusion also follows from his continuous Witt-vector action [18, Theorem 0.1]. We reproduce the concrete Steinberg-group arguments of the companion [13, Lemmas 6.4–6.7], since their exact identities, rather than identities of matrix images, are needed here.

4.1 Block roots and triangular evaluation

For the next three lemmas, let B be any unital $\mathbb{F}_2$ -algebra. Fix $n \geq 3$ and $b \geq 1$, and label nb coordinates by (i, s) , where $1 \leq i \leq n$ and $0 \leq s < b$. We call i the outer index and s the inner index. For $A = (a_{st}) \in \text{Mat}_b(B)$ and $i \neq j$, define the *block root*

$$X_{ij}(A) = \prod_{s,t=0}^{b-1} x_{(i,s),(j,t)}(a_{st}) \in \text{St}_{nb}(B).$$

All factors commute, because their source outer index is i and their target outer index is the different index j .

Lemma 4.1 (Block roots and coordinate changes). *The block roots satisfy the Steinberg relations over $\text{Mat}_b(B)$:*

$$\begin{aligned} X_{ij}(A + A') &= X_{ij}(A)X_{ij}(A'), \\ [X_{ij}(A), X_{kl}(A')] &= 1 && \text{if } j \neq k \text{ and } i \neq l, \\ [X_{ij}(A), X_{jk}(A')] &= X_{ik}(AA') && \text{if } i, j, k \text{ are distinct.} \end{aligned}$$

Consequently they define a homomorphism $\text{St}_n(\text{Mat}_b(B)) \longrightarrow \text{St}_{nb}(B)$. Moreover, for every $P \in \text{GL}_b(\mathbb{F}_2)$ there is an element $q \in \text{St}_{nb}(B)$ such that

$$q^{-1}X_{ij}(A)q = X_{ij}(P^{-1}AP) \quad (i \neq j).$$

Its matrix image is the block diagonal matrix with a copy of P in every outer block.

Proof. Additivity and the commuting relation follow factor by factor. For the consecutive commutator, the only nontrivial elementary commutators are

$$[x_{(i,s),(j,t)}(a_{st}), x_{(j,t),(k,v)}(a'_{tv})] = x_{(i,s),(k,v)}(a_{st}a'_{tv}).$$

Each resulting root commutes with both input blocks. Expanding the commutator and using additivity at each position therefore gives the entries $\sum_t a_{st}a'_{tv}$ of AA' .

An elementary operation supported inside outer block i acts on the outgoing strip to block j by the corresponding row operation; an operation inside block j acts by the inverse column operation. These are exact strip actions by Lemma 3.1. An operation inside any other outer block centralizes the block root. Gaussian elimination over $\mathbb{F}_2$ writes P as a product of elementary matrices: every nonzero pivot is 1, and a row transposition is a product of three row additions. Lift those factors inside each outer block. Applying the strip actions successively gives the asserted conjugation. For $b = 1$, $P = 1$ and the assertion is immediate. $\square$

If $v(t)$ is a root word over the polynomial algebra $B[t]$, with t central, and $T \in \text{Mat}_b(\mathbb{F}_2)$, let $W_v(T)$ be its evaluation at $t = T$, followed by the block-root homomorphism. The substitution $B[t] \rightarrow \text{Mat}_b(B)$ is a unital algebra homomorphism: the numerical matrix T commutes with every scalar matrix from B . In particular, if $v(t)$ has identity matrix image over $B[t]$, then $W_v(T)$ has identity matrix image over B . For each inner index s , also write

$$\iota_s : \text{St}_n(B) \longrightarrow \text{St}_{nb}(B), \quad x_{ij}(a) \longmapsto x_{(i,s),(j,s)}(a).$$

These coordinate homomorphisms have commuting images for distinct s ; we do not assume they are injective.

Lemma 4.2 (Triangular evaluation). *Let U be generated by the roots $x_{(i,s),(j,t)}(a)$ with $s < t$, allowing $i = j$. Then U injects into its matrix image and is normalized by every root whose inner indices agree. If a root word $v(t)$ over $B[t]$ has identity matrix image, and $T \in \text{Mat}_b(\mathbb{F}_2)$ is upper triangular with constant diagonal entry λ , then*

$$W_v(T) = \prod_{s=0}^{b-1} \iota_s(v(\lambda)). \quad (4.1)$$

Proof. The gap of a generator of U is $t - s > 0$. Two such roots either commute or have a commutator root whose gap is the sum of their gaps, by the consecutive-root relations in the two possible orders. They cannot be opposite roots. For $g \geq 1$, let U_g be generated by roots of gap at least g ; thus $U_1 = U$ and $U_b = 1$. Conjugating a generator of U_g by a generator of U adds only roots of larger gap, so U_g is normal in U . Modulo U_{g+1} , the group U_g is generated by the commuting roots of gap g .

It follows, successively for $g = 1, \dots, b - 1$, that every element of U is an ordered product with one factor at each coordinate pair, in increasing gap; repeated coefficients at a given pair are added. This is a finite collection process, since there are finitely many coordinate pairs and

$U_b = 1$. If its matrix image is the identity, the entries of gap one are precisely the corresponding collected coefficients, so they vanish. After coefficients of smaller gap have vanished, the same assertion holds at gap g : any product of two remaining entry matrices has gap greater than g . Induction makes every coefficient zero and proves faithfulness.

A root with equal inner indices cannot be opposite to a positive-gap root. Its commutator with a positive-gap root is either trivial or has the same positive gap. This proves that it normalizes U .

For a polynomial entry $a(t)$, the matrix $a(T)$ is upper triangular with diagonal entries $a(\lambda)$. Split its block root into its commuting diagonal and strictly upper factors. The diagonal factors normalize U , so the strictly upper factors in the complete word can be collected on the right. Diagonal factors at different inner indices commute, giving

$$W_v(T) = \left(\prod_s \iota_s(v(\lambda)) \right) u, \quad u \in U.$$

The left side and all the words $v(\lambda)$ have identity matrix image. Hence u has identity matrix image. Faithfulness of U gives $u = 1$ and proves (4.1). $\square$

4.2 Removing the wrap of a cyclic shift

Let C_b be the numerical cyclic shift from row s to column $s + 1$ modulo b , and let S_b be the shift with the same entries except for the entry from row $b - 1$ to column 0, which is removed. Thus S_b is strictly upper triangular. The next lemma replaces C_b by S_b without changing the evaluated Steinberg element. Its proof confines the difference to a faithful strip between two disjoint bands of inner indices.

Lemma 4.3 (Removing the wrap). *Let $v(t)$ be a product of $m \geq 1$ root factors over $B[t]$, whose polynomial entries have degrees at most $d \geq 0$. Suppose its matrix image is the identity over $B[t]$. Put $\delta = \max\{1, d\}$. If*

$$b > 2(m + 1)\delta,$$

then $W_v(C_b) = W_v(S_b)$ in $\text{St}_{nb}(B)$.

Proof. For each factor, split its cyclic evaluation as $D_h F_h$, where D_h is its evaluation at S_b and F_h consists of the entries that wrap past inner index $b - 1$. The splitting is exact because all roots in one outer rectangle commute. Since $d < b$, a term of degree $a \leq d$ goes from s to $s + a$ and can wrap only once. Every root in F_h therefore has source in $\{b - \delta, \dots, b - 1\}$ and target in $\{0, \dots, \delta - 1\}$.

For $r \geq 1$, set

$$H_r = \{b - r\delta, \dots, b - 1\}, \quad L_r = \{0, \dots, r\delta - 1\}.$$

When $b > 2r\delta$, the roots from inner indices in H_r to inner indices in L_r , with arbitrary outer indices, form an abelian strip V_r that injects into matrices by Lemma 3.1. Figure 1 displays these two bands. We shall show that conjugation by one D_h enlarges their widths by at most δ .

More precisely, for $r \leq m$ we claim that

$$D_h^{-1} V_r D_h \subseteq V_{r+1}. \tag{4.2}$$

Fix a root of V_r . The constituents of D_h all have the same outer source i and outer target $j \neq i$, and inner displacements between 0 and d . They commute with one another. A constituent that commutes with the initial root also commutes with every conjugate of it by the other constituents, and so can be omitted from this conjugation.

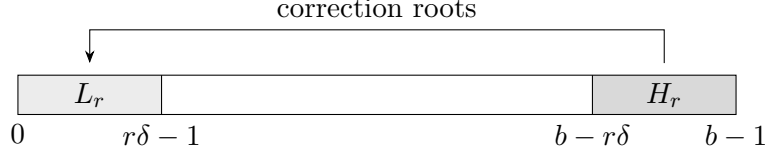


Figure 1: The correction strip V_r goes from the right band H_r to the left band L_r ; each band includes every outer index. The proof shows that one truncated block factor moves each endpoint by at most δ . The bound $b > 2(m+1)\delta$ keeps the enlarged bands disjoint throughout the collection.

An interacting constituent changes the source endpoint only when its target is the old source; the new source is at most d positions to the left. It changes the target endpoint only when its source is the old target; the new target is at most d positions to the right. These statements follow directly from the consecutive-root relations, including their reverse order. An opposite-root interaction cannot occur. Initially the source inner index minus the target inner index is greater than $b - 2r\delta$. After one endpoint has moved, the difference is still greater than

$$b - (2r + 1)\delta > \delta \geq d.$$

Thus no constituent, whose forward displacement is at most d , can be opposite to either the initial root or a root obtained after one endpoint move.

Nor can displacements accumulate repeatedly at one endpoint during this conjugation. A changed source has outer index i , whereas every constituent has target outer index $j \neq i$; the source cannot change again. A changed target has outer index j , whereas every constituent has source outer index i ; the target cannot change again. Both endpoints may change, but each changes at most once. A root with both endpoints changed has outer indices i, j and commutes with every constituent. All roots produced lie in V_{r+1} , proving (4.2). This reasoning applies to a product of strip roots as well.

It remains to collect the corrections without conjugating them by one another. Put $D_{>h} = D_{h+1} \cdots D_m$ and $D_{>m} = 1$. The exact group identity

$$\prod_{h=1}^m (D_h F_h) = (D_1 \cdots D_m) \prod_{h=1}^m (D_{>h}^{-1} F_h D_{>h}),$$

where the final product is in increasing order of h , does this. Every F_h starts in V_1 , and every suffix has at most m factors. By (4.2), all final corrections belong to V_{m+1} . Therefore

$$W_v(C_b) = W_v(S_b)u, \quad u \in V_{m+1}.$$

Both evaluated words have identity matrix image. The same is true of u , and the disjointness of H_{m+1} and L_{m+1} makes V_{m+1} faithful. Hence $u = 1$. $\square$

4.3 Graded evaluation and powers of central elements

Return to the graded algebra $R = \bigoplus_{a \geq 0} R_a$ with $R_0 = \mathbb{F}_2$. The map

$$\gamma : R \longrightarrow R[t], \quad \sum_a r_a \longmapsto \sum_a r_a t^a \quad (r_a \in R_a)$$

is a unital algebra homomorphism. Applying it to a root word representing $z \in J(R)$ gives a word $z(t)$ with identity matrix image over $R[t]$. Its evaluations at 1 and 0 are z and the degree-zero reduction of z , respectively.

For $T \in \text{Mat}_b(\mathbb{F}_2)$, fix an injection from the pairs (i, s) , where i ranges over the stable index set and $0 \leq s < b$, into the stable index set. Define $W_z(T)$ by applying the graded substitution and evaluating by block roots in those coordinates. This is a homomorphism on $\text{St}(R)$ by Lemma 4.1, so $W_z(T)$ depends only on z . The element $W_z(T)$ belongs to $J(R)$. All words and their block evaluations use finitely many coordinates; choosing a different injection gives the same element by centrality and Lemma 3.3.

Lemma 4.4 (Vanishing at long cyclic shifts). *If $z \in J(R)$ maps to 1 in $\text{St}(\mathbb{F}_2)$, then*

$$W_z(C_b) = 1 \quad \text{for every sufficiently large integer } b.$$

More precisely, if the graded word $z(t)$ has $m \geq 1$ factors of degrees at most d , it suffices that $b > 2(m+1)\max\{1, d\}$.

Proof. Choose one finite root word for z , and apply Lemma 4.3 to its graded word. Since S_b is strictly upper triangular, Lemma 4.2 gives, in the stable group,

$$W_z(C_b) = W_z(S_b) = \prod_{s=0}^{b-1} \iota_s(z(0)) = 1.$$

The fixed injection extends each ι_s to a homomorphism on the stable group. Since $z(0)$ is trivial there by hypothesis, every displayed coordinate image is trivial. The empty word satisfies the conclusion for all b . $\square$

Proposition 4.5 (Relative 2-primary torsion). *Every element of*

$$J(R) \cap \ker(\text{St}(R) \longrightarrow \text{St}(\mathbb{F}_2))$$

has finite 2-power order.

Proof. Let z be such an element. If b is a power of two, then $C_b^b = I$ and characteristic two gives $(C_b - I)^b = 0$. A basis adapted to the successive kernels of the nilpotent map $C_b - I$ makes C_b upper triangular with all diagonal entries 1. Thus some $P \in \text{GL}_b(\mathbb{F}_2)$ has this effect. By Lemma 4.1, conjugation by a Steinberg lift of the repeated basis change takes $W_z(C_b)$ to $W_z(P^{-1}C_bP)$. Centrality of $J(R)$ makes these elements equal. Lemma 4.2 then gives

$$W_z(C_b) = \prod_{s=0}^{b-1} \iota_s(z) = z^b. \tag{4.3}$$

The last equality uses Lemma 3.3: every coordinate image of the central element z equals z in the stable group. Choose a power of two b large enough for Lemma 4.4. Then $z^b = 1$. $\square$

Only the positive grading and the degree-zero field were used in this section. To detect central elements by finite degree truncations, we next use the additional phase structure of the specific algebra R . It will yield an odd-power counterpart of (4.3).

5 Commutators of long entries

The phase property will let us distribute a long word among several coordinate blocks. To do this inside the Steinberg group, we must control the extra factors produced by moving its roots. The main point of this section is that these factors lie in an abelian subgroup on which the matrix map is injective. Opposite roots require particular care: their commutation cannot be inferred just from their matrix images.

Fix an odd integer L and a threshold $N \geq 1$ from Lemma 2.3. Let k bound the lengths of the defining word rules, and fix

$$M \geq 2N + k + 1. \quad (5.1)$$

All groups in this section are stable groups over R . We identify a nonzero word class with its basis element in R , and write $|w|$ for its length. Write E_{ij} for the matrix unit in position (i, j) . Phases always take values in $\mathbb{Z}/L$.

5.1 Opposite roots and movement of a cut

Lemma 5.1. *Suppose $a, c \in R$ satisfy $ca = 0$. For distinct indices i, h, j and every $b \in R$,*

$$[x_{ij}(ab), x_{ji}(c)] = [x_{ih}(a), x_{hi}(bc)]. \quad (5.2)$$

In particular, $[x_{ij}(a), x_{ji}(c)]$ is independent of $j \neq i$. It commutes with both defining roots, has square 1, and has matrix image $I + acE_{ii}$.

Proof. Apply Lemma 3.4 to

$$X = x_{ih}(a), \quad Y = x_{hj}(b), \quad Z = x_{ji}(c).$$

Then $A = [X, Y] = x_{ij}(ab)$ and $B' = [Y, Z] = x_{hi}(bc)$. The Steinberg relations give $[A, X] = [B', Z] = 1$. They also give $[X, Z] = 1$ because $ca = 0$, and $[A, B'] = 1$ because $(bc)(ab) = b(ca)b = 0$. This proves (5.2).

Taking $b = 1$ proves independence of the auxiliary index. Express the commutator using an index h different from i, j . Each of its two factors then commutes with $x_{ij}(a)$ and with $x_{ji}(c)$: the only potential nonzero consecutive-root coefficient is ca . Thus the commutator commutes with both defining roots. Since each root has square 1, expanding $[x_{ij}(a)^2, x_{ji}(c)]$ shows that its square is 1. Finally, multiplication of the four elementary matrices, using $ca = 0$, gives $I + acE_{ii}$. $\square$

Lemma 5.2. *For each coordinate i and every nonzero word class w satisfying $|w| \geq M$ and $\sigma(w) \neq \tau(w)$, there is an element $H_i(w) \in \text{St}(R)$ with the following property. If p, q are nonzero words with*

$$|p|, |q| \geq N, \quad |p| + |q| \geq M, \quad \sigma(p) \neq \tau(q),$$

then, for every $j \neq i$,

$$[x_{ij}(p), x_{ji}(q)] = \begin{cases} H_i(w), & pq \text{ represents the nonzero class } w, \\ 1, & pq = 0 \text{ in } R. \end{cases} \quad (5.3)$$

The element $H_i(w)$ has square 1 and matrix image $I + wE_{ii}$.

Proof. The phase mismatch gives $qp = 0$, so Lemma 5.1 applies. We first show that the commutator does not change when the cut between p and q moves, provided both new parts still have length at least N .

For a move to the right, write the original word as abc , with $a = p$ and $bc = q$, and with $|c| \geq N$. The word c is nonzero and has $\tau(c) = \tau(q)$. Hence $ca = 0$, and (5.2), together with auxiliary-index independence, identifies the commutators for the cuts $a \mid bc$ and $ab \mid c$. This equality remains valid if $ab = 0$, in which case the commutator is 1. Otherwise $\sigma(ab) = \sigma(a)$, so the new cut retains the mismatch. For a move to the left, write $p = ab$ and $q = c$ with $|a| \geq N$. Now a is nonzero with $\sigma(a) = \sigma(p)$, so $ca = 0$ again, and the same identity compares $ab \mid c$ with $a \mid bc$. If the new suffix is zero we again obtain 1; otherwise its end phase is unchanged. This proves invariance under every allowed movement of the cut. If a part becomes zero, the equalities already obtained show that the original commutator was 1, and the argument can stop.

Next consider a single defining replacement inside a word of length $n \geq M$. The allowed cut positions are the integers in $[N, n - N]$. A rule of length at most k can straddle at most $k - 1$ of these positions, whereas (5.1) supplies more than k allowed positions. We can therefore move the cut so that the entire replacement lies in one part. That part has the same value in R before and after the replacement. If it is nonzero, its phases are unchanged because phases depend only on the word class. The other part is unchanged. The commutator consequently survives the replacement with the same value and mismatch, unless a zero part has already proved the original commutator to be 1.

If the concatenation is nonzero, every part of every equivalent word is nonzero. Cut movement and the preceding replacement argument show that the commutator depends only on the class of the concatenation. Conversely, a nonzero class w as in the statement can be cut into two parts of lengths at least N . The subword assertion of Lemma 2.3 makes their start and end phases $\sigma(w)$ and $\tau(w)$, respectively. Define $H_i(w)$ by any such cut. The independence just proved makes this definition unambiguous.

If the concatenation represents zero, the criterion (2.3) gives a sequence of replacements ending at a word that contains a zero-declared subword. Follow this sequence with the commutator. Either a part has already become zero, or the cut can be moved once more to put the final zero-declared subword wholly in one part. That part is zero in R , and the commutator is 1. This proves the zero case as well. The order and matrix-image assertions now follow from Lemma 5.1. $\square$

5.2 A subgroup detected by its matrix entries

Let S be a finite set of coordinates, with a phase $P(i) \in \mathbb{Z}/L$ assigned to each $i \in S$. An entry w from i to j will be called *consistent* if it is a nonzero word class of length at least M and

$$\sigma(w) = P(i), \quad \tau(w) = P(j).$$

We call it a *wrong-row entry* if instead

$$|w| \geq M, \quad \sigma(w) \neq P(i), \quad \tau(w) = P(j). \quad (5.4)$$

For $i \neq j$, a wrong-row entry defines the root $x_{ij}(w)$. For $i = j$, it defines $H_i(w)$ by Lemma 5.2. Let $U(S, P)$ be the subgroup generated by all these elements.

Lemma 5.3. *The subgroup $U(S, P)$ is abelian of exponent at most 2, and the matrix map is injective on it. Every root with a consistent entry normalizes $U(S, P)$.*

Proof. Write $U = U(S, P)$. First consider an off-diagonal generator $x_{ij}(a)$ of U and an off-diagonal root $x_{ef}(b)$ whose entry is either wrong-row or consistent. If $j = e$ and $i \neq f$, their commutator is $x_{if}(ab)$. The product is zero when b is wrong-row, since $\tau(a) = P(j) \neq \sigma(b)$. When b is consistent and $ab \neq 0$, the product inherits $\sigma(a)$ and $\tau(b)$, so it is a wrong-row entry from i to f . If $i = f$ and $j \neq e$, the reverse product ba is zero, because $\tau(b) = P(i) \neq \sigma(a)$, and the roots commute.

If the two positions are opposite, so that $e = j$ and $f = i$, the same mismatch $\sigma(a) \neq \tau(b)$ allows us to use (5.3). The commutator is $H_i(ab)$ when $ab \neq 0$, and is 1 when $ab = 0$. In the first case ab is a wrong-row entry from i to itself. In particular the commutator is 1 whenever b is wrong-row. All remaining pairs of positions commute by the defining Steinberg relations. This proves both the required commutations between off-diagonal generators of U and their required behavior under consistent roots.

Now take a diagonal generator $H_i(w)$. Represent it as a commutator through an auxiliary coordinate $h \notin S$. To compare it with a root $x_{ef}(b)$ on S , choose h outside its indices as well. If neither index is i , the supports are disjoint and the elements commute. If one index is i , the root lies in a strip between $\{i, h\}$ and its complement. By Lemma 3.1, conjugation by $H_i(w)$ acts on that strip by its matrix $I + wE_{ii}$. This matrix is its own inverse, since $w^2 = 0$ by the phase mismatch. Consequently, for $f \neq i$ and $e \neq i$,

$$\begin{aligned} H_i(w)x_{if}(b)H_i(w)^{-1} &= x_{if}(b + wb), \\ H_i(w)x_{ei}(b)H_i(w)^{-1} &= x_{ei}(b + bw). \end{aligned}$$

The incoming product bw is always zero, because $\tau(b) = P(i) \neq \sigma(w)$. The outgoing product wb is zero when b is wrong-row; when b is consistent and $wb \neq 0$, it is a wrong-row entry from i to f . These formulas prove that diagonal generators commute with off-diagonal generators of U , and that their commutators with consistent roots belong to U .

Two diagonal generators at different coordinates can be represented using disjoint pairs of coordinates, and hence commute. For two at the same coordinate i , write the second as

$$H_i(w') = [x_{it}(a), x_{ti}(c)], \quad ac = w', \quad |a|, |c| \geq N,$$

using t distinct from the auxiliary coordinate for the first. Here $\sigma(a) = \sigma(w') \neq P(i)$ and $\tau(c) = \tau(w') = P(i)$. The phase rule therefore gives $wa = 0$ and $cw = 0$. Applying the same strip action to the two displayed factors shows that $H_i(w)$ fixes both, and thus commutes with $H_i(w')$. The use of parts of length at least N is sufficient for these two applications of the phase rule.

We have proved that all generators of U commute. They have square 1 by root addition and Lemma 5.2. Their matrix images are $I + wE_{ij}$, indexed by positions and nonzero word classes satisfying (5.4). Any two added matrices multiply to zero: a potentially nonzero product requires matching positions, and then the end phase of the first entry is the designated row phase of the second, which differs from its start phase. Thus a product of distinct generators has matrix image

$$I + \sum_{(i,j,w)} wE_{ij}.$$

The word classes form a basis of R , so this is the identity only when the sum is empty. This proves injectivity on U . Finally, the commutator calculations show that conjugation by any consistent root sends every generator of U into U . Such a root is its own inverse, so it normalizes U . $\square$

6 Detection by degree truncations

We now show that degree truncations detect every element of $\text{St}(R)$. The central kernel is the essential case. For a representative whose entries are sufficiently long, phases turn cyclic-shift evaluation into an odd power of the original element. For an element invisible in all truncations, Lemma 4.4 makes the same evaluation trivial. Its already established 2-power order then forces the element itself to be trivial.

Keep L, N, M as in Section 5. For a root word with entries of lengths at least M , the following lemma puts the start and end phases directly into the coordinate labels.

Lemma 6.1 (Relocation by phases). *Suppose $z \in J(R)$ is represented by a finite word*

$$z = \prod_{\nu=1}^m x_{i_\nu j_\nu}(w_\nu), \quad |w_\nu| \geq M, \quad (6.1)$$

where every w_ν is a nonzero word class. Label new coordinates by (i, s) , with $s \in \mathbb{Z}/L$. In the stable group,

$$z = \prod_{\nu=1}^m x_{(i_\nu, \sigma(w_\nu)), (j_\nu, \tau(w_\nu))}(w_\nu). \quad (6.2)$$

Proof. The empty word presents no issue, so assume $m \geq 1$, and let S contain the finitely many original indices and at least three coordinates. Put $A = \mathbb{F}_2 \oplus R_{\geq M} \subseteq R$. We shall compare several evaluations of the word (6.1). Choose numerical columns $u_s \in \mathbb{F}_2^L$ and rows $v_s \in (\mathbb{F}_2^L)^*$ such that $v_s u_s = 1$ for every $s \in \mathbb{Z}/L$. On the word basis of the tail, set

$$\rho(w) = u_{\sigma(w)} v_{\tau(w)} w, \quad \rho(1) = I_L, \quad (6.3)$$

and extend linearly to A . This is a unital algebra homomorphism $A \rightarrow \text{Mat}_L(R)$. Indeed, a nonzero product ww' has matching middle phases and inherits the outside phases, so

$$\rho(w)\rho(w') = u_{\sigma(w)}(v_{\tau(w)}u_{\sigma(w')})v_{\tau(w')}ww' = \rho(ww').$$

If $ww' = 0$, the product of the two entry matrices is zero as well, regardless of the middle numerical pairing. Scalar products are immediate.

Applying ρ to the entries in (6.1), and then using the block roots of Lemma 4.1, produces an element of $J(R)$. To justify this assertion, the original matrix word is the identity already over the subalgebra A , since all its entries belong to A and $A \hookrightarrow R$ is injective. Its image under ρ is therefore the identity matrix. This uses only the matrix identity; no injectivity assertion about $\text{St}(A) \rightarrow \text{St}(R)$ is needed.

Write $\mathbf{e}_s$ for the standard numerical columns and $\mathbf{e}_s^*$ for their rows. Start with $u_s = \mathbf{e}_0$ and $v_s = \mathbf{e}_0^*$ for every s . The evaluated word is z supported on the coordinates $(i, 0)$, which is the same stable element by Lemma 3.3 and centrality. We change this evaluation in three steps.

Add entries in unused columns. Keep all $u_s = \mathbf{e}_0$, and change the rows to

$$v_0 = \mathbf{e}_0^*, \quad v_t = \mathbf{e}_0^* + \mathbf{e}_t^* \quad (t \neq 0).$$

Every pairing $v_t u_s$ is still 1. Each block root acquires at most one additional factor, going from a coordinate $(i, 0)$ to an unused coordinate (j, t) with $t \neq 0$. These factors belong to the strip from $S \times \{0\}$ to $S \times (\mathbb{Z}/L \setminus \{0\})$. The original factors, supported on $S \times \{0\}$, normalize this strip by Lemma 3.1. We can therefore collect the additional factors into a single strip element. Both the

old and new evaluated words have identity matrix image, so that strip element has identity matrix image. Strip injectivity makes it trivial. The new evaluation is still z .

Change the numerical basis. The rows v_t just defined form a basis. Let Q be the numerical matrix with these rows, so that $v_t Q^{-1} = \mathbf{e}_t^*$. Also $Q\mathbf{e}_0 = \sum_s \mathbf{e}_s$, because $v_t \mathbf{e}_0 = 1$ for every t . Conjugating all entry matrices by Q therefore changes the columns and rows to

$$u_s = \sum_{r \in \mathbb{Z}/L} \mathbf{e}_r \quad \text{for every } s, \quad v_t = \mathbf{e}_t^*.$$

Apply the exact block-conjugation assertion of Lemma 4.1 with $P = Q^{-1}$. This simultaneous basis change is induced by conjugation in $\text{St}(R)$. The evaluated word belongs to the central kernel, so it remains z .

Remove the factors in the wrong rows. Give coordinate (i, s) the designated phase $P((i, s)) = s$. A block root with entry w now factors as

$$\prod_{r \in \mathbb{Z}/L} x_{(i,r),(j,\tau(w))}(w).$$

Its factors commute. The factor with $r = \sigma(w)$ is the root in (6.2); all the other factors are wrong-row generators of $U(S \times \mathbb{Z}/L, P)$. Lemma 5.3 says that the retained, consistent roots normalize this subgroup. Collecting the other factors expresses the evaluated word as the right side of (6.2) times an element $u \in U(S \times \mathbb{Z}/L, P)$. The retained word itself has identity matrix image: apply (6.3) with $u_s = \mathbf{e}_s$ and $v_t = \mathbf{e}_t^*$. Hence u has identity matrix image. Injectivity on the wrong-row subgroup gives $u = 1$, proving (6.2). $\square$

Proposition 6.2. *Under the hypotheses of Lemma 6.1,*

$$W_z(C_L) = z^L. \quad (6.4)$$

Proof. For a homogeneous word w of degree $d = |w|$, the cyclic-shift evaluation of $x_{ij}(w)$ is the block root with entry wC_L^d . Since $\tau(w) - \sigma(w) = d$ in $\mathbb{Z}/L$, its factors are

$$x_{(i,\sigma(w)+a),(j,\tau(w)+a)}(w), \quad a \in \mathbb{Z}/L. \quad (6.5)$$

We claim that any two such factors arising from different offsets $a \neq a'$ commute, even when they come from different roots of the original word.

Write their entries as w, w' . A match between the column coordinate of the first factor and the row coordinate of the second requires $\tau(w) + a = \sigma(w') + a'$. Since $a \neq a'$, this implies $\tau(w) \neq \sigma(w')$, and hence $ww' = 0$. A match in the reverse direction similarly implies $w'w = 0$. Thus the Steinberg relations prove the claim unless the two roots are opposite. In that case both products vanish, and the reverse match also gives $\sigma(w) \neq \tau(w')$. Lemma 5.2, whose length hypotheses hold because $|w|, |w'| \geq M \geq N$, then gives $[x_{uv}(w), x_{vu}(w')] = 1$. This proves the claim in every case.

We may consequently collect the product by offsets, preserving within each offset the order of the original word. The offset-zero word is precisely the relocated word of Lemma 6.1, and every other offset word is its coordinate permutation. They all give z , since $z \in J(R)$. There are L offsets, proving (6.4). $\square$

It remains to obtain a long-entry representative from triviality in a degree truncation. Stable coordinates make this possible: a spare coordinate separates the two factors of a long word while it is being conjugated.

Lemma 6.3. *For every integer $M \geq 1$, each element of*

$$\ker(\mathrm{St}(R) \longrightarrow \mathrm{St}(R/R_{\geq 2M}))$$

can be represented by a finite product of roots whose entries are nonzero word classes of lengths at least M .

Proof. For any two-sided ideal K of a unital ring, the Steinberg presentation shows that the kernel of $\mathrm{St}(R) \rightarrow \mathrm{St}(R/K)$ is the normal closure of the roots $x_{ij}(a)$ with $a \in K$. Indeed, after these roots are killed, root addition makes $x_{ij}(a)$ depend only on the class of a modulo K , and the remaining relations are exactly the Steinberg relations over R/K . For $K = R_{\geq 2M}$, root addition expands every such entry in the homogeneous word basis. It is therefore enough to handle a conjugate $gx_{ij}(w)g^{-1}$ with $|w| \geq 2M$.

Choose a representative word $w = ab$ with $|a|, |b| \geq M$. Both parts are nonzero because w is nonzero. Let S contain i, j and every coordinate used by a fixed root word for g , and choose $h \notin S$. The consecutive-root relation gives

$$gx_{ij}(w)g^{-1} = [gx_{ih}(a)g^{-1}, gx_{hj}(b)g^{-1}].$$

The two conjugated factors lie respectively in the strips from S to $\{h\}$ and from $\{h\}$ to S . More explicitly, write $A_g = \pi(g)|_S$. By Lemma 3.1,

$$\begin{aligned} gx_{ih}(a)g^{-1} &= \prod_{r \in S} x_{rh}((A_g)_{ri}a), \\ gx_{hj}(b)g^{-1} &= \prod_{s \in S} x_{hs}(b(A_g^{-1})_{js}). \end{aligned}$$

Since $R_{\geq M}$ is a two-sided ideal, every displayed entry still belongs to $R_{\geq M}$. Expand the two strip elements into roots, expand each entry in the word basis, and then expand their commutator. All resulting root entries have length at least M . Doing this for the finitely many conjugates in a normal-closure expression proves the lemma. $\square$

Theorem 6.4. *Degree truncations detect the stable Steinberg group:*

$$\bigcap_{d \geq 1} \ker(\mathrm{St}(R) \longrightarrow \mathrm{St}(R/R_{\geq d})) = 1. \quad (6.6)$$

Proof. Let z lie in the displayed intersection. Its matrix image becomes the identity modulo every $R_{\geq d}$. The grading has $\bigcap_d R_{\geq d} = 0$, so its matrix image is the identity over R and $z \in J(R)$. Taking $d = 1$ also shows that z maps to 1 in $\mathrm{St}(\mathbb{F}_2)$.

By Lemma 4.4, $W_z(C_b) = 1$ for every sufficiently large integer b . Choose an odd phase modulus L from Lemma 2.3 beyond this threshold. This choice uses a fixed representative of z and precedes every choice of a long-entry representative. Now choose its phase threshold N and an integer $M \geq 2N + k + 1$.

The element z is trivial in the truncation at degree $2M$. Lemma 6.3 supplies a representative with nonzero word entries of lengths at least M . Applying Proposition 6.2 to this representative yields

$$z^L = W_z(C_L) = 1.$$

On the other hand, Proposition 4.5 gives $z^{2^a} = 1$ for some $a \geq 0$, since $z \in J(R)$ has trivial reduction to $\mathrm{St}(\mathbb{F}_2)$. The integers L and 2^a are coprime. Thus $z = 1$. $\square$

7 Integral stability in a fixed number of coordinates

The detection argument used additional coordinates in the stable Steinberg group. To apply Theorem 6.4 to a finitely presented group, we must now show that adjoining these coordinates does not kill an element already in a fixed rank. We prove that rank twelve suffices. All homology in this section has integral coefficients.

Throughout this Section, let $B = \mathbb{F}_2 \oplus I_B$ be a unital associative algebra, with I_B a two-sided ideal such that every finite square matrix over I_B is nilpotent. No grading or finite generation assumption is needed. We often suppress B from the notation for its matrix and Steinberg groups.

Proposition 7.1 (Finite-rank inputs). *For such an algebra B , the following statements hold.*

- (i) *Every tuple of columns of B^m with linearly independent reductions in $\mathbb{F}_2^m$ extends to an invertible matrix.*
- (ii) *The kernel $J_r(B)$ is central in $\text{St}_r(B)$ for $r \geq 6$.*
- (iii) *For $r \geq 5$, the group $E_r(B)$ is normal in $\text{GL}_r(B)$, and the natural maps*

$$\text{GL}_r(B)/E_r(B) \longrightarrow \text{GL}_{r+1}(B)/E_{r+1}(B)$$

are isomorphisms.

These are, respectively, Lemma 5.1, Proposition 5.4, and Lemma 5.5 of the companion paper [13]. They have exactly the hypotheses stated above. The integral conclusion we need is the following.

Theorem 7.2. *Let $B = \mathbb{F}_2 \oplus I_B$ satisfy the hypotheses of Proposition 7.1. For every $n \geq 12$, the standard stabilization map $J_n(B) \rightarrow J_{n+1}(B)$ is an isomorphism, and the natural homomorphism*

$$\text{St}_n(B) \longrightarrow \text{St}(B)$$

is injective.

The ring B is local: its elements outside I_B are units, whereas its elements in I_B are nilpotent. Thus the injectivity conclusion also follows, with a better rank bound, from the classical stability theorem of Suslin and Tulenbaev [16, Theorem 4.1]. We include a direct proof to make the integral homology comparison and its natural maps explicit. It compares successive ranks by the action on ordered frames, following the classical method of homological stability for linear groups; see van der Kallen [17]. Two details matter here: we use ordered frames with integral chains, and identify the second homology of their stabilizers by universal central extensions. We give both arguments explicitly.

7.1 Central extensions of frame stabilizers

A group is *perfect* if it equals its commutator subgroup. A *universal central extension* of a perfect group H is a central extension $\tilde{H} \rightarrow H$ admitting a unique homomorphism over H to every central extension of H . Its kernel is naturally $H_2(H; \mathbb{Z})$; this identification is natural for homomorphisms lifted between universal central extensions. We use this standard connection between Steinberg groups and second homology in its integral form; see [19, Theorems III.5.4 and III.5.5].

For $r \geq 6$ and $l \geq 0$, put

$$S_{r,l} = \left\{ \begin{pmatrix} A & 0 \\ C & I_l \end{pmatrix} : A \in E_r(B), C \in \text{Mat}_{l \times r}(B) \right\}. \quad (7.1)$$

This is the semidirect product of its additive lower-left strip with $E_r(B)$. Replace $E_r(B)$ in this semidirect product by $\text{St}_r(B)$, acting through its matrix image, and denote the resulting group by $\tilde{S}_{r,l}$. There is a natural surjection $\tilde{S}_{r,l} \rightarrow S_{r,l}$ with kernel $J_r(B)$.

Lemma 7.3. *The map $\tilde{S}_{r,l} \rightarrow S_{r,l}$ is a universal central extension. Consequently the inclusion of the complementary block induces a natural isomorphism*

$$H_2(E_r(B); \mathbb{Z}) \xrightarrow{\sim} H_2(S_{r,l}; \mathbb{Z}), \quad (7.2)$$

and both groups identify naturally with $J_r(B)$.

Proof. The extension is central by Proposition 7.1(ii), because its kernel acts trivially on the strip. Both its source and target are perfect. Indeed, an internal root is a consecutive-root commutator, and a strip root has the same expression using an intermediate index among the first r coordinates.

Consider an arbitrary central extension $T \rightarrow S_{r,l}$. The roots available in $S_{r,l}$ are $e_{ij}(a)$ with $j \leq r$ and $i \neq j$, where i may also be one of the last l coordinates. Choose arbitrary lifts $y_{ij}(a) \in T$. We first show that lifts of roots appearing in a Steinberg commuting relation actually commute:

$$[y_{ij}(a), y_{kt}(b)] = 1 \quad \text{if } j \neq k \text{ and } i \neq t. \quad (7.3)$$

Choose $h \leq r$ distinct from i, j, k, t . In matrices,

$$e_{ij}(a) = [e_{ih}(a), e_{hj}(1)],$$

and both factors on the right commute with $e_{kt}(b)$. In T , their commutators with $y_{kt}(b)$ are therefore central. Conjugating each factor by $y_{kt}(b)$ changes it only by a central factor, so fixes their commutator. Since this commutator differs from $y_{ij}(a)$ by a central element, (7.3) follows. The bound $r \geq 6$ supplies every spare index used below as well.

Define corrected lifts by

$$\hat{y}_{ij}(a) = [y_{ih}(a), y_{hj}(1)], \quad h \leq r, \quad h \neq i, j. \quad (7.4)$$

They are independent of h and satisfy

$$[y_{ih}(a), y_{hj}(b)] = \hat{y}_{ij}(ab) \quad (h \leq r; \ i, h, j \text{ distinct}). \quad (7.5)$$

To prove this, choose a fourth distinct index $f \leq r$ and apply Lemma 3.4 to

$$X = y_{ih}(a), \quad Y = y_{hf}(b), \quad Z = y_{fj}(1).$$

All its commuting hypotheses follow from (7.3), since $[X, Y]$ and $[Y, Z]$ lift, respectively, $e_{if}(ab)$ and $e_{hj}(b)$. The resulting identity equates the left side of (7.5) with $[y_{if}(ab), y_{fj}(1)]$: replacing a lift by a central multiple does not change a commutator. Setting $b = 1$ proves independence in (7.4); the same identity then proves (7.5) for arbitrary b .

The corrected lifts are additive in their parameters. In (7.4), use $y_{ih}(a)y_{ih}(a')$ as a lift of $e_{ih}(a + a')$ and expand its commutator with $y_{hj}(1)$. Equation (7.3) allows the commutator factors to pass the ih factors, giving

$$\hat{y}_{ij}(a + a') = \hat{y}_{ij}(a)\hat{y}_{ij}(a').$$

Here $\hat{y}_{ij}(0) = 1$, since any lift of an identity matrix is central. Together with (7.3) and (7.5), this proves all Steinberg relations for the corrected lifts on the available roots.

On the first r coordinates they therefore define a homomorphism $\text{St}_r(B) \rightarrow T$. On the strip they define a homomorphism from the additive group $\text{Mat}_{l \times r}(B)$ to T . These homomorphisms respect the semidirect-product action: conjugation of a strip root by an internal root is determined by a commuting relation, except when their indices form a consecutive path, in which case it is determined by (7.5). These are exactly the matrix conjugation rules. Thus they give a homomorphism $\tilde{S}_{r,l} \rightarrow T$ over $S_{r,l}$. It is unique, since two such homomorphisms differ by a homomorphism from the perfect group $\tilde{S}_{r,l}$ to the central, abelian kernel of $T \rightarrow S_{r,l}$. This proves universality.

Taking $l = 0$ identifies $H_2(E_r(B); \mathbb{Z})$ with $J_r(B)$. The inclusion $E_r(B) \rightarrow S_{r,l}$ lifts by adjoining the zero strip and is the identity on the kernels $J_r(B)$. Naturality of the universal central extension now proves (7.2). $\square$

7.2 The integral complex of ordered frames

An *ordered frame* of B^m is an ordered tuple of columns whose reductions modulo I_B are linearly independent. Let $C_p(m)$ be the free abelian group on ordered frames of length $p + 1$, for $p \geq 0$, with boundary the alternating sum of deletions. Set $C_{-1}(m) = \mathbb{Z}$, generated by the empty frame, and augment by sending each one-column frame to 1. Every ordering is a separate generator; we impose no relation that identifies two orderings by a sign.

The corresponding finite complexes are complexes of injective words on matroid independence complexes. Their shellability is established in [7, Theorem 1.2 and Example 3.3]. The following direct proof gives the precise integral vanishing range we need.

Lemma 7.4. *The augmented chain complex $C_\bullet(m)$ is exact in degrees at most $m - 2$.*

Proof. First let V be a finite labeled collection of nonzero vectors spanning a vector space of dimension m . Different labels remain different vertices even if their vectors agree. Write $C_\bullet(V)$ for the augmented complex on its ordered independent tuples. We prove the claimed vanishing by induction on the number of labels, allowing $m = 0$: the empty collection has its only chain group in degree -1 , and the asserted range is then empty.

Delete a label v , and let $V' = V \setminus \{v\}$. In the relative complex

$$Q_\bullet = C_\bullet(V)/C_\bullet(V'),$$

the generators are the independent tuples containing v . Filter $Q_\bullet$ increasingly by the number a of labels before v . Deleting v is zero in the relative complex; deleting a label before it lowers the filtration; deleting a label after it preserves the filtration. Thus the associated graded complex for a fixed a is a direct sum, over independent ordered prefixes $(u_1, \dots, u_a, v)$, of complexes in which only the suffix varies.

For one such prefix, project all remaining labeled vectors to the quotient by $\langle u_1, \dots, u_a, v \rangle$, and discard the zero projections. The projected collection spans dimension $m - a - 1$ and has fewer labels than V . Its augmented ordered-tuple complex, shifted up by $a + 1$ degrees, is precisely the corresponding summand of the associated graded complex, up to the constant boundary sign $(-1)^{a+1}$. By induction its homology vanishes through degree

$$(m - a - 1) - 2 + (a + 1) = m - 2.$$

This also covers a zero-dimensional quotient: its sole homology in degree -1 shifts to degree $a = m - 1$, outside the asserted range. The filtration is finite, so $H_i(Q_\bullet) = 0$ for $i \leq m - 2$.

If V' still spans dimension m , induction and the long exact sequence of the pair give the required vanishing for V . Otherwise V' spans dimension $m - 1$, and v is independent of its entire span. Prepending v to every tuple of V' gives a chain homotopy from its inclusion into $C_\bullet(V)$ to zero; in augmented degree -1 it sends 1 to (v) . The induced maps $H_i(C_\bullet(V')) \rightarrow H_i(C_\bullet(V))$ are therefore zero. They are also surjective for $i \leq m - 2$, by the relative vanishing just proved. The homology of $C_\bullet(V)$ vanishes in this range.

Finally, any cycle in $C_\bullet(m)$ uses finitely many columns. Adjoin the standard basis columns to this finite collection. Their reductions, with column labels retained, form a finite collection spanning $\mathbb{F}_2^m$. The finite result supplies a bounding chain in the asserted range, proving the Lemma. $\square$

7.3 Second homology in successive ranks

Proof of Theorem 7.2. We now combine the stabilizer calculation with the ordered-frame complex to prove Theorem 7.2. Fix $m \geq 13$ and let $E_m(B)$ act on ordered frames by left multiplication. For frame lengths $l \leq 5$, this action is transitive. To see this, complete a given frame to an invertible matrix, placing it in the last l columns, and put $r = m - l$. By Proposition 7.1(iii), the map $GL_r / E_r \rightarrow GL_m / E_m$ is an isomorphism. Right multiplication by a suitable complementary GL_r block therefore brings the completion into E_m without changing its last l columns.

The pointwise stabilizer of the last l standard columns is exactly $S_{m-l,l}$ from (7.1). Indeed, its general form has $A \in GL_{m-l}$ and arbitrary lower-left strip. The strip is elementary, and the injectivity of $GL_{m-l} / E_{m-l} \rightarrow GL_m / E_m$ says that this matrix belongs to E_m precisely when $A \in E_{m-l}$. Write $S_l = S_{m-l,l}$ for the rest of the proof. All these complementary ranks are at least eight, so Lemma 7.3 applies.

Take a free right $\mathbb{Z}[E_m]$ -resolution $P_\bullet \rightarrow \mathbb{Z}$ of the trivial module and form the first-quadrant double complex

$$D_{p,q} = P_q \otimes_{\mathbb{Z}[E_m]} C_p(m), \quad p, q \geq 0. \quad (7.6)$$

This is the usual equivariant homology double complex; see [1, Chapter VII]. Its total homology in degree two is $H_2(E_m; \mathbb{Z})$ via augmentation. For completeness, adjoining the augmentation column $p = -1$ gives the same tensor product with the augmented frame complex. Each P_q is free and hence flat, so Lemma 7.4 makes the rows exact in the stated range. The augmented total complex has zero homology in degrees one and two. The long exact sequence separating its augmentation column from (7.6) consequently gives the asserted isomorphism.

Take vertical homology first. For $0 \leq p \leq 4$, the permutation module on ordered $(p+1)$ -frames has one orbit and stabilizer S_{p+1} . Shapiro's Lemma identifies the first page as

$$E_{p,q}^1 = H_q(S_{p+1}; \mathbb{Z}), \quad 0 \leq p \leq 4. \quad (7.7)$$

The first differential is the alternating sum of the face maps, each expressed by a stabilizer inclusion and a coordinate change that returns the remaining frame to the chosen standard frame.

The row $q = 1$ is zero in these columns, because every S_l is perfect. In row $q = 0$, all entries are $\mathbb{Z}$ and each face map is the identity. Hence the first differential is zero for odd p and the identity for even p . In particular,

$$E_{1,1}^2 = E_{2,1}^2 = E_{2,0}^2 = E_{3,0}^2 = 0. \quad (7.8)$$

In row $q = 2$, the two face maps from column one to column zero coincide. Indeed, the inclusion $E_{m-2} \rightarrow S_2$ induces an isomorphism on H_2 by Lemma 7.3, so it suffices to compare the faces on this complementary block. The first standard frame is (e_{m-1}, e_m) . Deleting its first column

gives e_m . Deleting its second column gives e_{m-1} , which is returned to e_m by swapping the last two coordinates. This swap is elementary in characteristic two and centralizes the first $m-2$ coordinates. The two induced maps on that block are therefore identical. Their alternating sum vanishes, and

$$E_{0,2}^2 = H_2(S_1; \mathbb{Z}).$$

No higher differential leaves position $(0, 2)$. The only possible incoming differentials are from $(2, 1)$ on the second page and $(3, 0)$ on the third page; both vanish by (7.8). The other positions of total degree two, $(1, 1)$ and $(2, 0)$, also vanish. The spectral sequence thus shows that the edge map

$$H_2(S_1; \mathbb{Z}) \longrightarrow H_2(E_m; \mathbb{Z})$$

is an isomorphism. This is the map induced by inclusion: the zero-th column corresponds to the standard column e_m , and its map into the total complex followed by augmentation is the usual homology map for its stabilizer inclusion.

Combining with (7.2), we have proved that the natural stabilization map

$$H_2(E_{m-1}(B); \mathbb{Z}) \longrightarrow H_2(E_m(B); \mathbb{Z})$$

is an isomorphism for $m \geq 13$. The natural identifications by universal central extensions turn this into the isomorphism $J_{m-1}(B) \rightarrow J_m(B)$.

Finally, an element killed by $\text{St}_n(B) \rightarrow \text{St}(B)$ already has identity matrix image, since coordinate embeddings of matrix groups are injective. It therefore lies in $J_n(B)$. If $n \geq 12$, the successive isomorphisms on these kernels show that it cannot become trivial at any later rank. The stable group is the direct limit, so this proves the injectivity asserted in Theorem 7.2. $\square$

8 The finitely presented residually finite 2-group

Stable detection and the fixed-rank injectivity just proved now yield finite quotients separating the group we will use. The ring R throughout this section is the algebra of Proposition 2.1, with the hierarchy of Proposition 2.2. Its finite degree quotients will give actual finite groups separating the elements of $\Gamma = \text{St}_{12}(R)$.

Proposition 8.1. *For every $d \geq 1$, the group $\text{St}_{12}(R/R_{\geq d})$ is finite. The homomorphisms*

$$\text{St}_{12}(R) \longrightarrow \text{St}_{12}(R/R_{\geq d}), \quad d \geq 1,$$

separate the elements of $\text{St}_{12}(R)$.

Proof. Put $B_d = R/R_{\geq d}$. This ring is finite: it is the direct sum of finitely many finite-dimensional vector spaces over $\mathbb{F}_2$. Moreover $B_d = \mathbb{F}_2 \oplus I_d$ with $I_d^d = 0$, so every finite square matrix over I_d is nilpotent. The centrality assertion in Proposition 7.1 therefore applies to B_d . The kernel of

$$\text{St}_{12}(B_d) \longrightarrow \text{E}_{12}(B_d)$$

is central, and the matrix group on the right is finite. Hence the center of $\text{St}_{12}(B_d)$ has finite index. Schur's theorem states that a group with center of finite index has finite derived subgroup [15]; see also [14, Theorem 10.1.4]. Since $\text{St}_{12}(B_d)$ is perfect, it is itself finite.

Let $g \neq 1$ in $\text{St}_{12}(R)$. Theorem 7.2 makes its image in $\text{St}(R)$ nonidentity, and Theorem 6.4 detects that image in $\text{St}(B_d)$ for some d . The natural square

$$\begin{array}{ccc} \text{St}_{12}(R) & \longrightarrow & \text{St}(R) \\ \downarrow & & \downarrow \\ \text{St}_{12}(B_d) & \longrightarrow & \text{St}(B_d) \end{array}$$

commutes on each root generator. Thus the image of g in $\text{St}_{12}(B_d)$ is nonidentity. No injectivity assertion for the bottom arrow is needed. This proves separation by the finite groups just constructed. $\square$

Proof of Theorem 1.1. The algebra R is infinite and finitely presented as a unital $\mathbb{F}_2$ -algebra. Accordingly $\Gamma = \text{St}_{12}(R)$ has an ordinary finite presentation by [13, Theorem 4.1]; this is also a case of [8, Theorem 3]. The group Γ is infinite, since its matrix image contains distinct matrices $e_{12}(a)$ for all $a \in R$. It is residually finite by Proposition 8.1.

Define

$$G = \ker(\Gamma \longrightarrow \text{St}_{12}(\mathbb{F}_2)),$$

where the map is induced by the degree-zero projection of R . The case $d = 1$ of Proposition 8.1 shows that this target is finite. Thus G has finite index in Γ . It is infinite and has an ordinary finite presentation by the Reidemeister–Schreier theorem [9, Chapter II, Section 4]. Restricting the separating finite-quotient maps of Γ proves residual finiteness of G .

Fix $g \in G$. Its matrix image is $I + A$ for some $A \in M_{12}(I)$. Proposition 2.1 gives an integer $r \geq 1$ with $A^r = 0$. Choose $a \geq 0$ with $2^a \geq r$. In characteristic two,

$$(I + A)^{2^a} = I + A^{2^a} = I.$$

Consequently $g^{2^a} \in J_{12}(R)$, and its reduction to $\text{St}_{12}(\mathbb{F}_2)$ is still trivial. Its stable image belongs to $J(R)$ and reduces to 1 in $\text{St}(\mathbb{F}_2)$, so Proposition 4.5 kills that image by a 2-power 2^b . The injectivity in Theorem 7.2 then yields $g^{2^{a+b}} = 1$ in Γ , hence in G . This proves every assertion of Theorem 1.1. $\square$

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